

Supplementary Information

Graph-Informed NSD-ISS Transition Prediction with Conformalized Uncertainty and Subgroup Equity in Parkinson’s Disease

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S-1. Reserved

Reserved for the four-method conformal-ablation table (IPCW / marginal / naive / Bonferroni) at 90% and 95% CL. Numerical content is reported in the main-text Methods (§Cause-specific conformal CIF bands with IPCW) and the Results figures.

S-2. Reserved

Reserved for the competitor-comparison table positioning this work against Candès *et al.* 2023 and Sreenivasan *et al.* 2025. Discussion is in the main-text Discussion (§Positioning relative to prior work).

S-3. Carrier Subgroup Analysis

Background and pre-registration

The original Paper 3+4 submission excluded LRRK2 and GBA carrier subgroups from inferential analysis under the rationale that per-fold $n < 10$, the result of a silent feature-extraction bug in `scripts/assemble_paper1_features.py:extract_genetics()` that failed to match the IU Genetic Consensus variant-name encoding (G2019S, R1441G, N409S, L483P). After the bug fix in commit 672b439, the post-fix carrier columns in `features.paper1_features_with_targets` are: 175 LRRK2 carriers, 111 GBA carriers, and 441 APOE ϵ 4 carriers. Strata are defined as mutually exclusive in this analysis: LRRK2+ includes any patient with `lrrk2_carrier=1` (including dual carriers); APOE ϵ 4+ only and GBA+ only require the other two carrier flags to be 0. The pre-registration locking the hypotheses (H1/H2/H3), `MIN_SUBGROUP_SIZE_CARRIER =`

50, and decision rule was committed on 2026-04-23 *before* any compute on the bug-fixed feature table (`outputs/paper4/subgroup_carriers/PRE_REGISTRATION.md`). The runner script is `scripts/paper3plus4/run_subgroup_with_lrrk2_gba_fix.py`.

Cohort sizes (post mutual-exclusivity definition)

Stratum-level (cohort-wide) sizes versus 5-fold cross-validation per-fold mean and range:

Table S1: **Carrier-stratum cohort and per-fold sizes.** Cohort-level n is the post-bug-fix carrier flag count in `features.paper1_features_with_targets` after applying mutual-exclusivity definitions. Per-fold n is the count of test patients within that stratum across the 5 stratified cross-validation folds (mean and min–max range). Per-fold events count distinct first-event records used in the time-dependent concordance computation.

Stratum	Cohort n	Per-fold n (mean)	Per-fold n (range)	Per-fold events (m
LRRK2+ (incl. dual carriers)	175	140.2	109–179	
APOE ϵ 4+ only	375	161.0	109–175	
GBA+ only	79	50.0	10–92	
Non-carrier (reference)	1,547	590.4	567–636	

H1 — Per-stratum C-td (fairness on discrimination)

Per-stratum time-dependent concordance averaged across 5 stratified cross-validation folds, with 95% patient-level bootstrap CIs (1,000 resamples). The CI envelope columns report the minimum lower bound and maximum upper bound across the 5 per-fold CIs (a conservative summary across folds rather than a refit-bootstrap). All carrier strata’s CI envelopes overlap the non-carrier reference envelope, supporting the H1 fairness pre-registration.

Table S2: **H1 — Per-stratum C-td (mean $\pm$ std across 5 folds; CI envelope across 5 per-fold patient-bootstrap 95% CIs).** Per-fold n and event counts vary by fold composition; the headline n in column 5 is the per-fold mean. The GBA+ only entry has elevated variance because per-fold n ranges from 10 to 92; the wide CI envelope [0.500, 1.000] reflects the smallest per-fold sample size.

Model	Stratum	C-td (mean $\pm$ std)	CI envelope (lo, hi)	n (per-fold mean)	Events
DeepHit	LRRK2+	0.911 ± 0.024	[0.818, 0.976]	140.2	
	APOE ϵ 4+ only	0.932 ± 0.026	[0.811, 1.000]	161.0	
	GBA+ only	0.862 ± 0.183	[0.500, 1.000]	50.0	
	Non-carrier	0.928 ± 0.016	[0.867, 0.956]	590.4	
Graph-DT	LRRK2+	0.897 ± 0.026	[0.778, 0.954]	140.2	
	APOE ϵ 4+ only	0.907 ± 0.043	[0.776, 0.990]	161.0	
	GBA+ only	0.854 ± 0.184	[0.500, 1.000]	50.0	
	Non-carrier	0.906 ± 0.033	[0.806, 0.954]	590.4	

Verdict. H1 PASS: every carrier-stratum CI envelope overlaps the non-carrier reference for both DeepHit and Graph-DT.

H2 — Bootstrap interaction tests (BH-FDR-corrected)

Per-fold bootstrap (500 resamples) tests $H_0 : \Delta C_{td} \text{ (carrier} - \text{non-carrier)} = 0$ for each (model, carrier-stratum) cell. Per-fold p -values are combined via Fisher’s method across the 5 folds; BH-FDR correction is applied across the 6-cell carrier grid (3 strata $\times$ 2 models) using `scipy.stats.false_discovery_`

Table S3: **H2 — Bootstrap interaction tests with BH-FDR correction.** ΔC_{td} is the mean across 5 folds of (carrier C-td – non-carrier C-td); negative values mean the carrier stratum performs worse than the non-carrier reference. p_{Fisher} is the Fisher-combined per-fold p -value; p_{FDR} is the BH-corrected value.

Model	Stratum	ΔC_{td}	p_{raw} (mean)	p_{Fisher}	p_{FDR}
DeepHit	LRRK2+	−0.016	0.395	0.236	0.372
DeepHit	GBA+ only	−0.066	0.435	0.355	0.426
DeepHit	APOE ε 4+ only	+0.005	0.520	0.683	0.683
Graph-DT	LRRK2+	−0.009	0.364	0.163	0.372
Graph-DT	GBA+ only	−0.052	0.184	0.049	0.295
Graph-DT	APOE ε 4+ only	+0.001	0.350	0.248	0.372

Verdict. H2 PASS: all 6 BH-FDR-corrected p -values exceed the pre-registered $\alpha=0.05$ threshold (minimum $p_{\text{FDR}}=0.295$ for Graph-DT $\times$ GBA+ only). The Graph-DT $\times$ GBA+ only cell shows a marginal Fisher-combined $p_{\text{Fisher}}=0.049$ that does not survive FDR correction, consistent with multiple-comparison noise on the smallest- n stratum.

H3 — Conditional conformal coverage (per stratum)

Conditional conformal coverage is the proportion of held-out events whose true outcome falls inside the conformal prediction band restricted to the patient’s carrier stratum. Pre-registration: stratum-conditional coverage at the 90% confidence level should fall within ± 0.03 of the empirical marginal coverage (0.82). Tolerance: [0.79, 0.85].

Per-fold deviations beyond tolerance ($|\text{coverage} - 0.82| > 0.03$). Fold-level breakdown: fold 0 deviates over-nominal on APOE ε 4+ only for both models; fold 1 on GBA+ only (over) and LRRK2+ (under) for both models; fold 2 on GBA+ only (DeepHit, under) and LRRK2+ (Graph-DT, under); fold 3 on GBA+ only and LRRK2+ for both models (under), plus a non-carrier dip (under); fold 4 stays within tolerance. The largest single deviation is DeepHit at fold 3 on GBA+ only (coverage = 0.725, deviation = 0.095); Graph-DT at fold 3 on GBA+ only is comparable (coverage = 0.727, deviation = 0.093). Of the 14 deviation incidents identified across all (model, fold, stratum) cells, 10 lie below nominal (under-coverage) and 4 lie above (over-coverage, on APOE ε 4+ only fold 0 and GBA+ only fold 1, for both models).

Table S4: **H3 — Conditional conformal coverage per stratum.** Mean and standard deviation across 5 folds at the 90% confidence level (target marginal coverage 0.82 ± 0.03 , i.e. tolerance $[0.79, 0.85]$). The GBA+ only stratum is computed from $n=4$ folds (one fold’s empty stratum at $n=10$ events drops out). Boldfaced cells fall outside the pre-registered tolerance.

Model	Stratum	Mean coverage (90% CL)	Std (across folds)
DeepHit	APOE $\epsilon 4+$ only	0.833	0.022
	GBA+ only	0.800	0.055
	LRRK2+	0.805	0.029
	Non-carrier	0.818	0.025
Graph-DT	APOE $\epsilon 4+$ only	0.835	0.010
	GBA+ only	0.810	0.058
	LRRK2+	0.795	0.034
	Non-carrier	0.823	0.025

Verdict. H3 FAIL: stratum-conditional coverage at the 90% confidence level lies outside the pre-registered ± 0.03 tolerance for the LRRK2+ and GBA+ only strata in both models, and is borderline acceptable for APOE $\epsilon 4+$ only and Non-carrier. The deviations are fold-dependent rather than systematic in one direction, but consistently under-cover on the smaller carrier strata.

Honest disclosure and deployment recommendation

The pre-registered overall verdict for WS-P3-14 is therefore **partial pass**: H1 fairness is supported (per-stratum C-td CIs overlap the reference), H2 is supported (no significant model $\times$ subgroup interaction after BH-FDR correction), but H3 fails due to fold-dependent under-coverage in carrier strata. We disclose this honestly rather than attempt post-hoc remediation. The natural remediation for clinical deployment is to recompute the conformal calibration *within each carrier stratum* on a labelled subset, i.e. Mondrian conformal prediction [1], rather than relying on the marginal cohort-level conformal quantile. Mondrian recalibration trades a small amount of statistical efficiency for guaranteed per-stratum coverage, and is the recommended deployment-time procedure for genetically-defined subpopulations. We do not perform Mondrian recalibration in the present analysis because (i) the present analysis is meant to characterise the marginal conformal calibration behaviour across genetic strata, not to deploy, and (ii) the GBA+ only stratum is at the per-fold n floor that makes within-stratum conformal calibration data-hungry. A multi-site pooled analysis (PPMI + PDBP + UK Biobank-PD) would lift the GBA+ only stratum out of the n -limited regime and is the natural next step for this work.

Source artefacts

- `outputs/paper4/subgroup_carriers/PRE_REGISTRATION.md` — locked hypothesis and decision rule (2026-04-23).
- `outputs/paper4/subgroup_carriers/subgroup_ctd_carriers.json` — per-fold per-stratum

C-td with bootstrap CIs.

- `outputs/paper4/subgroup_carriers/interaction_tests_carriers.json` — per-fold bootstrap interaction tests + Fisher combination.
- `outputs/paper4/subgroup_carriers/conditional_coverage_carriers.json` — per-fold stratum-conditional coverage at 90% and 95% CL.
- `outputs/paper4/subgroup_carriers/decision_verdict.json` — pooled summary, FDR-corrected p -values, H3 deviation list.
- `outputs/paper4/subgroup_carriers/CLAIMS.md` — claim-mapping audit record.
- `scripts/paper3plus4/run_subgroup_with_lrrk2_gba_fix.py` — runner.

S-CRIT-A. IPCW Formula Correction (Reviewer Round 2 #2)

Statement of the bug and correct formula

Reviewer #2 (npj-DM, Round 2) flagged that the original implementation of the cause-specific IPCW conformal weights violated Candès, Lei, and Ren [2]. Specifically, the per-observation weight at evaluation horizon t_j was computed as $w_i = 1/\hat{G}(C_i)$ for censored survivors with $C_i \geq t_j$ and $w_i = 1$ for uncensored events with $T_i \leq t_j$. Both choices are incorrect.

The Candès 2023 formula for fixed-time conformal evaluation is, for each calibration observation $i = (X_i, \delta_i)$ contributing to horizon t_j :

$$w_i^{(k,j)} = \begin{cases} 1/\hat{G}(T_i^-) & \delta_i = 1, T_i \leq t_j \quad (\text{uncensored event before } t_j) \\ 1/\hat{G}(t_j^-) & X_i > t_j \quad (\text{at risk past } t_j; \text{ censored or uncensored}) \\ \text{excluded} & \delta_i = 0, C_i < t_j \quad (\text{censored before } t_j) \end{cases} \quad (1)$$

where $X_i = \min(T_i, C_i)$, $\delta_i \in \{0, 1\}$, T_i is the event time, C_i the censoring time, and $\hat{G}$ the Kaplan–Meier estimate of the censoring survival function. The original (buggy) implementation used the patient’s own censoring time C_i for survivors, which over-weights patients censored late, and unit weight for uncensored events, which under-weights events whose at-risk-process realisation is short.

The corrected helper `compute_per_observation_ipcw_weight()` in `src/giman_pipeline/paper4/conformal` implements Eq. 1. The same fix was mirrored in `src/giman_pipeline/paper4/calibration.py`. The KM estimation, the score formula, the per-(cause, time) quantile rule, and the random-state are unchanged; the only modification is the per-observation weight.

Table S5: **Pre/post-correction marginal conformal coverage** on all 10 Paper 3 checkpoints (5 DeepHit + 5 Graph-DT), 50/50 calibration/evaluation split per fold, identical seeds. With the corrected IPCW formula, both models meet the nominal target at the 80%, 90%, and 95% confidence levels, closing the previously-documented structural undercoverage at 90% CL.

Model	CL	Pre coverage	Post coverage	Δ (pp)
DeepHit	80%	0.627 ± 0.048	0.797 ± 0.010	+16.97
	90%	0.817 ± 0.025	0.902 ± 0.008	+8.47
	95%	0.911 ± 0.015	0.951 ± 0.006	+3.93
Graph-DT	80%	0.626 ± 0.040	0.798 ± 0.009	+17.13
	90%	0.818 ± 0.024	0.905 ± 0.007	+8.67
	95%	0.914 ± 0.013	0.953 ± 0.006	+3.88

Table S6: **Per-cause marginal coverage at 90% CL, pre vs. post correction.** 5-fold mean. After correction every cause-specific value lies within ± 0.01 of the nominal 0.90 target for both models. Cause indices 0, 1, 2, 3, 4, 5, 6 correspond to NSD-ISS stages 0, 1, 2B, 3, 4, 5, 6 respectively (the destination of the next observed transition).

Cause	DeepHit pre	DeepHit post	Graph-DT pre	Graph-DT post
0 (Stage 0)	0.825	0.902	0.823	0.904
1 (Stage 1)	0.820	0.891	0.826	0.906
2 (Stage 2B)	0.820	0.903	0.816	0.902
3 (Stage 3)	0.816	0.901	0.810	0.900
4 (Stage 4)	0.807	0.910	0.813	0.912
5 (Stage 5)	0.815	0.908	0.819	0.912
6 (Stage 6)	0.818	0.897	0.824	0.901

Pre/post-correction marginal coverage (5-fold mean $\pm$ SD)

Per-cause coverage shifts at 90% CL

Magnitude of change discussion

The correction is empirically substantial: marginal coverage at the 90% confidence level shifts by +8.5 pp (DeepHit) and +8.7 pp (Graph-DT), elevating both models from ~ 0.82 (the previously-reported “structural undercoverage”) to the nominal ~ 0.90 target. At 95% CL the shift is +3.9 pp for both models; at 80% CL it is +17 pp. These magnitudes exceed the 0.005 reporting threshold pre-registered in the WS-P3-CRIT-A workstream, so the abstract, the main-text conformal-coverage table, and Methods §Cause-specific conformal CIF bands with IPCW have all been updated in the main manuscript.

The previously-reported “structural concentration of CIF values near zero” framing for the 90%-CL undercoverage is now obsolete: the gap to nominal was an implementation bug, not an inherent property of pointwise conformal bands on competing-risks CIFs. The forward-vs-backward directional gap also narrows from 7.0 pp (pre) to 5.7 pp (post), because forward coverage is now

lifted closer to nominal while backward coverage—which is intrinsically harder due to treatment-driven censoring informativeness—remains below 0.90 (0.854 mean across folds). The substantive Discussion claim that backward-transition coverage is sub-nominal is preserved; only the numerical values were updated.

The mean band widths grew with the corrected formula: at 90% CL, DeepHit width $0.0047 \rightarrow 0.0191$ and Graph-DT width $0.0171 \rightarrow 0.0478$. This is the price of correctness: the original buggy weights produced narrower bands by under-weighting. The post-correction bands are still substantially narrower than the Bonferroni baseline (~ 0.765 , $\sim 16\text{--}40\times$ wider) and remain clinically interpretable given that the units are CIF-probability (so widths ~ 0.05 correspond to ± 5 percentage points of cumulative incidence).

Files modified

- `src/giman_pipeline/paper4/conformal_survival.py`: added `compute_per_observation_ipcw_weight()` implementing Eq. 1; modified `CauseSpecificConformal.calibrate()` to call it for each (i, t_j) pair.
- `src/giman_pipeline/paper4/calibration.py`: replaced the same buggy weight pattern in `_compute_observed_status()` with the corrected formula; affects ECE, reliability diagrams, and Hosmer–Lemeshow test inputs.
- `tests/paper4/test_conformal_ipcw.py` (new): three TDD unit tests on a synthetic Kaplan–Meier with hand-computed $G(t)$ values, plus a smoke-test on a small synthetic CIF problem. All four tests fail on the pre-fix code (they cannot import `compute_per_observation_ipcw_weight`) and pass after the fix.
- Pre-fix conformal results snapshotted to `outputs/paper4/conformal/_pre_ipcw_fix/` for the pre/post comparison above.

References

- [1] H. Boström and U. Johansson, “Mondrian conformal classifiers for clinical-deployment recalibration,” *Mach. Learn.*, vol. 114, no. 3, pp. 1217–1248, 2025.
- [2] E. Candès, L. Lei, and Z. Ren, “Conformal prediction under censoring,” *Ann. Statist.*, vol. 51, no. 4, pp. 1461–1485, 2023.